Subhag Sharma

Roll No.: 2025PCS0042
M.Tech - Computer Science and Engineering
Indian Institute Of Technology, Jammu

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Github | Website
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Education

•Indian Institute of Technology Jammu *2025 – 2027 (Expected)*
M.Tech in Computer Science and Engineering *CGPA: 9.14 (Current)* Jammu, India

•Shri Mata Vaishno Devi University *2020 – 2024*
B.Tech in Computer Science and Engineering *CGPA: 8.91* Jammu, India

•Delhi Public School *2020*
Class XII *Percentage: 93%* Jammu, India

•Delhi Public School *2018*
Class X *Percentage: 93.4%* Jammu, India

Experience

•NIMS University *Jun 2024 – Aug 2024*
Research Associate Jaipur, Rajasthan

– Assisted and carried out research work under the MARIK School of Artificial Intelligence and Machine Learning, an initiative by the Czech Government and NIMS, focusing on Medical Images.
– Assisted with training on the latest technologies for the larger benefit of the students.

•Indian Institutes of Technology, Ropar *Apr 2024 – Jun 2024*
Intern Ropar, Punjab

– Engaged in cutting-edge medical image processing and segmentation research at IIT Ropar, an institution of national importance, contributing to innovative advancements in healthcare technology.
– Committed to pushing boundaries and delivering impactful solutions in an evolving learning environment.

•Jammu & Kashmir e-Governance Agency (JaKeGA) *Oct 2023 – Mar 2024*
Intern Jammu, J&K

– Created an RBSK (Rashtriya Bal Swasthya Karyakram) database & application as Team Lead for 2.7M students in 29K schools, revolutionizing healthcare with centralized records.
– Engineered a proprietary OCR software to streamline transformation of scanned official documents into text, enhancing accessibility for the disabled community and increasing document digitization by 50%.

Projects

•Explainable AI for Intestinal Parasite Detection
Interpretable Artificial Intelligence| Quantus, PyTorch

– Orchestrating a rigorous comparative evaluation between standard black-box feature extractors (e.g., MobileViT) and inherently interpretable paradigms, including Prototypical Part Networks (ProtoPNet), B-cos architectures, and Concept Bottleneck Models (CBM).
– Benchmarking model faithfulness and visual grounding in low spatial frequency domains, enforcing strict parameter randomization sanity checks to guarantee that diagnostic attributions are axiomatic, structurally sane, and robustly tied to parasitic morphological biomarkers.

•Multiple Sclerosis Lesion Segmentation in Brain MRI
Using SOTA U-Net Architectures with Deep Feature Extraction Backbone | TensorFlow, Keras, OpenCV, PyDicom, PIL

– This novel work aids in early diagnosis and treatment of this rare neurological disease, potentially enhancing quality of life for affected individuals.
– Performed comprehensive evaluation of various U-Net and deep feature extraction backbone combinations (ResNet, DenseNet) for MS lesion segmentation in brain MRI, identifying optimal configurations to enhance diagnostic precision.

Publications

- **Analysis of EEG Signals for Deceit Identification Using Various Supervised Learning Methods**
  Published in Scopus Indexed LNNS by Springer | DOI: 10.1007/978-981-97-0641-9_12
  - Discusses the role of Machine Learning in analyzing EEG to use brain signals for lie detection, with results at par with established physical formats of interrogation.
- **Early Detection of Alzheimer's Disease Using Advanced Machine Learning Techniques – A Comprehensive Review**
  Published in Scopus Indexed SIST | DOI: 10.1007/978-981-99-5180-2_37
  - Establishes prognostic measures for Alzheimer's Disease (affecting 73% of population aged 75+) using multiple modalities as inputs for state-of-the-art ML models.
- **Intelligent Diagnosis of Autism Spectrum Disorder: Harnessing ML for Enhanced Early Detection**
  Published in Scopus Indexed LNNS by Springer (IC3S Proceedings) | DOI: 10.1007/978-981-97-2550-2_27
  - Developed a standardized questionnaire accessible via a mobile application, with background algorithms trained on a vast dataset for ASD detection.
- **Revolutionizing Breast Cancer Detection: A Shallow Neural Network Approach for Classification in Mammographic Scans**
  Published in Scopus Indexed Handbook of Deep Learning Models for Healthcare Data Processing | DOI: 10.1201/9781003467281
  - Highlights the role of lightweight CNNs in comparison with heavy models, beating standardized benchmark results for calcification and mass classification.
- **Plant Leaf Disease Classification Using Knowledge Distillation Methodologies**
  Presented at CVIP 2024 |DOI: 10.1007/978-3-031-93697-5_11
  - Developed lightweight CNNs (as co-author/guide) for deployment in remote, resource-constrained areas to enable early detection and treatment of plant diseases in agrarian-dependent countries like India.

Technical Skills

- **Languages:** C, C++, Python, Java, SQL
- **Technologies:** TensorFlow, PyTorch, Keras, Pandas, Scikit-Learn, AWS, Google Cloud Platform, LaTeX
- **Concepts:** Compiler Design, Operating Systems, Virtual Memory, Cache Memory, Encryption/Decryption, Artificial Intelligence, Machine Learning, Neural Networks, Database Management, Cloud Computing

Certifications

- **Cloud:** Google Cloud Infrastructure: Core Services; Google Cloud Fundamentals: Core Infrastructure; Essential Google Cloud Infrastructure: Foundation
- **Machine Learning:** Supervised Machine Learning: Regression and Classification; Advanced Learning Algorithms; Unsupervised Learning, Recommender Systems & Reinforcement Learning; AI for Medicine

Achievements

- Got awarded the Reliance Foundation Postgraduate Scholarships, awarded to only 100 PG scholars across India *2025*
- Finished **Second** in AgriThon, an Agriculture Innovation Hackathon organized by SKUAST. *2025*
- All India Rank **911** in GATE CS/IT 2024 out of 1,70,825 candidates. *2024*
- Awarded Certificate of Distinction for being the best student in Academics, Co-Curricular & Extra-Curricular activities for the batch of 2024. *2024*
- Stood **First** in Cybersecurity Hackathon HackDefy organized by the Government of J&K. *2023*

Positions of Responsibility & Social Engagements

- **ML Co-Lead,** Google Developer Student Council (GDSC)
- **Organising Committee Member,** International Conference on Computing, Communications, and Cyber-Security
- **Organiser,** Two Faculty Development Programs with speakers from prestigious institutes in India and abroad on Image Processing and Machine Learning
- **Volunteer,** National Social Service (NSS)